

INTEGER HEIGHT-TWO RECTANGLE VISIBILITY GRAPHS ARE OUTERPLANAR

ABSTRACT. Caughman, Dunn, Laison, Neudauer and Starr conjectured that a graph of rectangle visibility height 2 is outerplanar. We prove it in their model, in which every rectangle has integer corners. The proof runs through a structure theorem: in any layout of integer rectangles inside a box of height 2, the visibility graph is exactly the union of two paths — one along the rectangles meeting the lower unit row, in x -order, one along those meeting the upper unit row — together with the pairs (bottom rectangle, top rectangle) whose open x -intervals overlap. Splitting each full-height rectangle into halves exhibits the graph as a contraction of a two-layer non-crossing drawing, whence outerplanarity. Integrality cannot be dropped: we exhibit a K_4 layout whose y -rescaling has bounding-box height exactly 2 with the same visibility graph.

1. THE CONJECTURE AND THE RESULT

A *rectangle visibility representation* (RV-representation) of a graph G is a family $\{R_v\}_{v \in V(G)}$ of closed axis-parallel rectangles $R_v = [x_1^v, x_2^v] \times [y_1^v, y_2^v]$ with pairwise disjoint interiors — shared edges are allowed — such that $uv \in E(G)$ if and only if R_u and R_v are *visible*, meaning that, after possibly exchanging u and v , either

- (h) $x_2^u \leq x_1^v$, and some interval J of positive length lies in $(y_1^u, y_2^u) \cap (y_1^v, y_2^v)$ such that for every $y \in J$ the open segment $(x_2^u, x_1^v) \times \{y\}$ meets no rectangle of the family; or
- (v) the same with the two axes exchanged.

This is the “axis-parallel band of positive width meeting no other closed rectangle” of [1], written out. Two rectangles sharing part of an edge are visible along it, the open segment then being empty. A graph with such a representation is a *rectangle visibility graph* (RVG); for the origins of the class see [4]. Following Caughman, Dunn, Laison, Neudauer and Starr [1, §2] we consider only *integer* rectangles, all of whose corners are lattice points; $\text{height}(G)$ is the least height of the bounding box of an integer RV-representation of G , and $F_{h,w}$ is the set of graphs realisable inside an $h \times w$ box.

Item 6 of the “Directions for Further Research” of [1] reads, verbatim,

- (1) *We conjecture that if $\text{height}(G) = 2$ then G is outerplanar. Is this true?*

We answer (1) affirmatively.

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Theorem 1. *If a graph G has an integer RV-representation whose bounding box has height 2, then G is outerplanar. Equivalently, in the model of [1], $\text{height}(G) = 2$ implies that G is outerplanar.*

The proof rests on a structure theorem. Fix a layout S of integer rectangles with bounding box $[0, w] \times [0, 2]$. Every $R \in S$ has integer $0 \leq y_1 < y_2 \leq 2$, so its y -interval is one of exactly three:

$$\text{BOTTOM} = [x_1, x_2] \times [0, 1], \quad \text{TOP} = [x_1, x_2] \times [1, 2], \quad \text{FULL} = [x_1, x_2] \times [0, 2].$$

Put $B^* = \text{BOTTOM} \cup \text{FULL}$ (the rectangles meeting the open strip $0 < y < 1$) and $T^* = \text{TOP} \cup \text{FULL}$.

Theorem 2 (Structure). *Members of B^* have pairwise disjoint open x -intervals, hence are linearly ordered by x ; likewise T^* . With those orders, the visibility graph of S is exactly*

$$\begin{aligned} E = & \text{path}(B^* \text{ in } x\text{-order}) \cup \text{path}(T^* \text{ in } x\text{-order}) \\ & \cup \{bt : b \in \text{BOTTOM}, t \in \text{TOP}, \mathring{I}_b \cap \mathring{I}_t \neq \emptyset\}, \end{aligned}$$

where $\mathring{I}_R = (x_1, x_2)$ is the open x -interval of R . In particular a FULL rectangle has no vertical visibility edge at all.

The two-row observation behind Theorem 2 appears in the source, which uses it in its own case analysis, and in tree-restricted form as Lemma 1 of the authors' earlier abstract [2]; we use it with the contraction argument of §3.

Remark 3 (integrality). Theorem 1 is false in the real-coordinate model. The four rectangles

$$[0, 2] \times [0, 1], \quad [0, 2] \times [2, 3], \quad [1, 2] \times [1, 2], \quad [2, 3] \times [0, 3]$$

have pairwise disjoint interiors inside a 3×3 box and their visibility graph is K_4 (all six pairs are visible; the reader may check the six bands directly). Apply $y \mapsto \frac{2}{3}y$:

$$[0, 2] \times [0, \frac{2}{3}], \quad [0, 2] \times [\frac{4}{3}, 2], \quad [1, 2] \times [\frac{2}{3}, \frac{4}{3}], \quad [2, 3] \times [0, 2].$$

An axis-parallel scaling is a homeomorphism preserving interiors, axis-parallel bands and positive width, so the visibility graph is still K_4 — and the bounding box now has height exactly 2. The integer model is the source's own, so the hypothesis costs nothing against the question as asked, but it must be stated.

2. PROOF OF THE STRUCTURE THEOREM

Lemma 4 (half-integer sight lines suffice). *Let A, B be members of a layout of integer rectangles. The set of abscissae at which a vertical segment joining the interiors of A and B meets no other rectangle of the layout is*

$$U = \mathring{I}_A \cap \mathring{I}_B \setminus \bigcup_C [x_1^C, x_2^C],$$

the union running over the rectangles C whose y -interval meets the open interval strictly between A and B . Every endpoint occurring in U is an integer, so each connected component of U is an open interval with integer endpoints. Hence $U \neq \emptyset$ if and only if $U \supseteq (c, c+1)$ for some integer c , if and only if $c + \frac{1}{2} \in U$ for some integer c . The same holds with the two axes exchanged. For a layout with arbitrary real coordinates the same argument gives the endpoints of each component of U among the finitely many x -coordinates of the layout, so $U \neq \emptyset$ if and only if U contains the midpoint of two consecutive distinct such coordinates.

Proof. U is an open set obtained from an open interval with integer endpoints by deleting finitely many closed intervals with integer endpoints, so its components are open intervals with integer endpoints; a nonempty such component contains a unit interval $(c, c+1)$ and therefore its midpoint. For the general statement, replace “integer” by “a coordinate of the layout” throughout: the deleted intervals and the ambient interval still have all their endpoints in that finite set, so each component of U does too, and a nonempty component therefore contains two consecutive distinct coordinates’ midpoint. $\square$

Proof of Theorem 2. Two members of B^* both contain the whole strip $\{0 < y < 1\}$ over their x -range, so if their open x -intervals met, their interiors would meet. Hence B^* is linearly ordered by x , and likewise T^* .

Horizontal edges. By Lemma 4 (axes exchanged) a horizontal band of positive width may be taken at $y = \frac{1}{2}$ or $y = \frac{3}{2}$. The rectangles meeting $y = \frac{1}{2}$ are exactly those of B^* , and each of them meets that line in its whole x -interval; so along that line they are linearly ordered and A sees B if and only if they are consecutive. That is the path along B^* in x -order. The line $y = \frac{3}{2}$ gives the path along T^* symmetrically.

Vertical edges. By Lemma 4 a vertical band may be taken at $x = c + \frac{1}{2}$ for an integer c . The rectangles meeting that line have pairwise disjoint y -interiors inside $[0, 2]$, and each is BOTTOM, TOP or FULL; so there are exactly three exhaustive cases: a FULL rectangle meets the line, in which case it is the only rectangle meeting it and no edge arises (whence a FULL rectangle has no vertical edge); or no FULL does and both a BOTTOM b and a TOP t meet it, in which case they are the only two, are consecutive, and bt is an edge; or at most one rectangle meets the line and no edge arises. Finally, “some integer c has $[c, c+1] \subseteq I_b \cap I_t$ ” is precisely $\min(x_2^b, x_2^t) > \max(x_1^b, x_1^t)$, i.e. $\dot{I}_b \cap \dot{I}_t \neq \emptyset$. $\square$

3. PROOF OF THEOREM 1

Recall that a graph is outerplanar if and only if its vertices admit a cyclic order in which the edges, drawn as chords of a circle, pairwise do not cross; and that outerplanarity is minor-closed.

Proof of Theorem 1. Let S realise G inside $[0, w] \times [0, 2]$.

Step 1: split the FULL rectangles. Replace each $F = [x_1, x_2] \times [0, 2]$ in S by the two halves $F_b = [x_1, x_2] \times [0, 1]$ and $F_t = [x_1, x_2] \times [1, 2]$. Interiors remain pairwise disjoint, so the result S' is a legal FULL-free layout in the same box; call its visibility graph G' . By Theorem 2, $B^*(S') = \text{BOTTOM} \cup \{F_b : F \in \text{FULL}\}$ carries the same bottom path as before, $T^*(S') = \text{TOP} \cup \{F_t : F \in \text{FULL}\}$ the same top path, and the only new overlap edges are the pairs $F_b F_t$: a pair (b, F_t) with $b \in \text{BOTTOM}$ is excluded because b and F both lay in $B^*(S)$ and so have disjoint open x -intervals, a pair (F_b, t) symmetrically, and a pair (F_b, F'_t) because F and F' both lay in $B^*(S)$. Each $F_b F_t$ is an edge of G' , since F_b and F_t have equal, hence overlapping, open x -intervals. Contracting all of them merges each pair $\{F_b, F_t\}$ back into F , leaves every other vertex alone, and returns exactly G . So G is a contraction, hence a minor, of G' .

Step 2: G' is outerplanar. Let $b_1 < \dots < b_p$ be the bottom rectangles of S' in x -order and $t_1 < \dots < t_q$ the top ones, and take the cyclic order

$$b_1, b_2, \dots, b_p, t_q, t_{q-1}, \dots, t_1.$$

Every vertex of G' occurs exactly once, since S' is FULL-free. The two paths of Theorem 2 run along consecutive positions of this cyclic order, so they cross nothing. Suppose two overlap edges crossed: $b_i t_j$ and $b_{i'} t_{j'}$ with $i < i'$ and $j' < j$. Then

$$\begin{aligned} x_2(b_i) &> x_1(t_j), & x_2(t_{j'}) &> x_1(b_{i'}), \\ x_2(b_i) &\leq x_1(b_{i'}), & x_2(t_{j'}) &\leq x_1(t_j), \end{aligned}$$

the last two because bottoms, respectively tops, have disjoint open x -intervals in x -order. Chaining gives $x_1(b_{i'}) < x_2(t_{j'}) \leq x_1(t_j) < x_2(b_i) \leq x_1(b_{i'})$, a contradiction. So no two chords cross and G' is outerplanar.

Step 3: conclude. G is a minor of G' by Step 1, G' is outerplanar by Step 2, and outerplanarity is minor-closed. Hence G is outerplanar. $\square$

4. STARS, AND THE FAILURE OF THE CONVERSE

Lemma 5. *Let S be a layout of integer rectangles in a box of height 2, G its visibility graph, and v a vertex whose neighbourhood is independent. Then $\deg v \leq 4$; and if R_v is not FULL then $\deg v \leq 3$.*

Proof. If R_v is FULL it lies on both paths of Theorem 2 and has no vertical edge, so $\deg v \leq 2 + 2 = 4$ with no hypothesis at all. Let R_v be BOTTOM (the TOP case is symmetric); it has at most two neighbours on the bottom path, so it suffices to show that at most one TOP rectangle overlaps it. Suppose $t < t'$ in T^* -order both have $\dot{I}_t \cap \dot{I}_v \neq \emptyset \neq \dot{I}_{t'} \cap \dot{I}_v$, and choose them consecutive among the members of T^* overlapping $\dot{I}_v$. Since $\dot{I}_v$ is an interval containing a point below $x_2(t)$ and a point above $x_1(t')$, it contains $[x_2(t), x_1(t')]$, hence the open x -interval of every member of T^* lying between t and t' . Such a member cannot be FULL, since a FULL rectangle lies in B^* together with R_v and so has open x -interval disjoint from $\dot{I}_v$; so it is a TOP overlapping $\dot{I}_v$,

contradicting the choice of t, t' as consecutive among those. Therefore t and t' are consecutive in T^* and tt' is an edge of the top path — but $t, t' \in N(v)$, contradicting independence. $\square$

Corollary 6. *If $K_{1,m} \in F_{2,w}$ for some w then $m \leq 4$, and if $m = 4$ the centre is realised by a FULL rectangle. In particular $K_{1,5}$ is outerplanar and has no integer RV-representation of bounding-box height 2: the converse of Theorem 1 fails.*

That the converse fails is not new: $K_{1,5}$ has six vertices, so it is one of the 112 rows of the authors' published census [3] of connected six-vertex graphs, where its entry reads $\text{height}(K_{1,5}) = 3$. What Lemma 5 adds is a reason valid for every m and every width at once, together with the FULL-centre clause at $m = 4$.

The bound is attained, by the source's own published layout of $K_{1,4}$ [1]:

$$(2) \quad \begin{aligned} a &= [1, 2] \times [1, 2], & b &= [3, 4] \times [1, 2], & c &= [2, 3] \times [0, 2], \\ d &= [0, 1] \times [0, 1], & e &= [4, 5] \times [0, 1]. \end{aligned}$$

Here $B^* = \{d, c, e\}$ and $T^* = \{a, c, b\}$ in x -order, and no BOTTOM overlaps a TOP, so Theorem 2 returns $E = \{ac, bc, cd, ce\}$: the star centred at the FULL rectangle c , inside the box $[0, 5] \times [0, 2]$.

5. THE ACCOMPANYING PROGRAM

Every object above is printed in full, so nothing in §§1–4 requires a program. One accompanies this note nonetheless: `verify.py`, with a recorded run in `verify.output.txt`. It uses the Python standard library only and exact integer and `Fraction` arithmetic, with no floating-point decision anywhere.

It implements the definition of visibility twice — once specialised to integer coordinates by testing unit columns and unit rows, once for arbitrary rational coordinates by testing the midpoint of every pair of consecutive distinct coordinates, which Lemma 4 shows to be exact — and checks that the two agree. It re-derives from the definition the visibility graphs of (2) and of the two layouts of Remark 3. It then enumerates every layout of integer rectangles whose bounding box is exactly $[0, w] \times [0, 2]$ for $w \leq 7$ — 711,202 layouts — and on each one verifies Theorem 2 edge-for-edge and re-runs Steps 1–3 of §3: the split layout's own definition-derived graph, the non-crossing property of the two-layer cyclic order, and that contracting the split pairs returns the original graph. Over those layouts the realised star sizes are $m \in \{1, 2, 3, 4\}$, the $m = 4$ centre is always FULL, and the bound of Lemma 5 is attained but never exceeded.

The program reports **VERDICT: ALL 45 CHECKS PASS**. What it does not do is stated in its own output: the enumeration stops at $w = 7$, and it is the proof of §§2–3, not the enumeration, that covers all w and all n ; nothing there verifies $\text{height}(K_{1,5}) = 3$, only that no height-2 layout in its range realises $K_{1,5}$; and no attribution in this note is machine-checked.

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